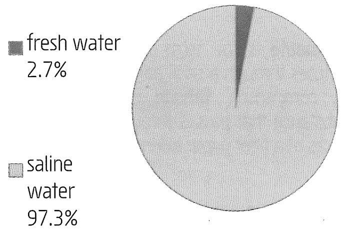
16.9 Need a drink?

On average, water makes up around 55-78% of a person's body mass, though this varies with gender, age and body weight. In our bodies, water plays a vital role as a solvent for metabolic processes, for transporting nutrients and is essential for the excretion of cellular waste from our body. Evaporation of water from our skin also helps regulate body temperature. It is said we can go a few weeks without food but only days without water.

Figure 17 Water on the Earth's surface.



Of all the water on Earth **97.3%** is **saline** with 97.2% of it in the ocean. Only **2.7%** occurs as **fresh water**. Most of this (68%) is tied up in the polar ice caps and glaciers with only **0.6 %** of the total available as fresh water in rivers, lakes, swamps or groundwater.

Although water is the most abundant compound on the Earth's surface finding some to drink is not so easy. (See Fig 17.) Most of the water around us is not suitable for drinking or cooking, as it contains dissolved salts which would dehydrate us or dissolved substances like organic compounds or heavy metals which may be toxic to our organs or carcinogenic (cancer causing). Even freshwater supplies may not be potable if they contain suspended matter such as clay particles or microorganisms (pathogens) that can cause diseases like cholera or dysentery.

A good understanding of chemical principles and methods of analysis are vital in designing and maintaining water treatment facilities for the production of water that is safe to drink and cook with. Water like this is known as **potable water**. Potable water should be clear, colourless, odourless, contain no pathogens or toxic substances and be pleasant to taste. In Australia the National Health and Medical Research Council (**NHMRC**) provides guidelines for water quality management. While there is no specific limit on the maximum allowable total dissolved solids (**TDS**) in drinking water, the NHMRC recommends 'for good palatability TDS in drinking water should not exceed 600 mg L^{-1}'. Such dissolved substances are typically in the form of various non-toxic ions, eg Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, SO_4^{2-}, HCO_3^-, CO_3^{2-}, SiO_4^{2-}, F^-, Fe^{3+}, Mn^{2+}, NO_3^-, NO_2^-, PO_4^{3-} or non-toxic dissolved organic matter.

Depending upon the source of water chosen it must be treated differently to make it potable. The traditional source of water in Western Australia has been rivers and dams. Water from these sources typically only required disinfection and fluoridation. However, due to increased demand and changing rainfall patterns, groundwater has become most significant and more recently drinking water is being sourced from the Indian Ocean. Water from these sources requires more intensive treatment to make it potable.

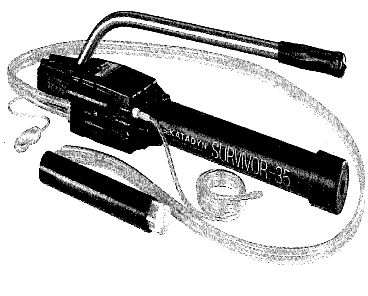
Perth, including the South West, the Goldfields and Agricultural regions share a supply system known as the **Integrated Water Supply System** (**IWSS**). In 2012-2013 the IWSS provided approximately 285 billion litres of fresh, clean drinking water to more than 1.9 million people. The mix of water sources for that year was: 40% groundwater, 27% seawater desalination from Kwinana plus a second major desalination plant at Binningup and surface water provided 33%. The proportion of desalinated water has now increased to well above 40% as the second stage of the 'Southern Seawater Desalination Plant' (Binningup plant) is now fully commissioned.

Attempt Set 28 # 12 and 13.

16.10 Potable water from seawater

The most abundant water source on Earth is sea water. Its high salinity, typically greater than 35,000 mg L^{-1} TDS, make it unsuitable for drinking, washing or agriculture. Two approaches used to produce potable water from sea water are **distillation** and **reverse osmosis**. (See Fig 18.) The principle of sea water distillation is the same as that applied in the laboratory (p25). Water, a volatile solvent, is evaporated from sea water, that contains salt, a non-volatile solute. The resulting water vapour is condensed to produce distilled water free of any dissolved salts. Usually the evaporation is carried out in several stages at progressively lower pressure, called multi-stage flash distillation, so that less energy is needed in the evaporation process. Also by using a heat exchange process the heat released from the condensation of water vapour is transferred to incoming cold sea water. This increases energy efficiency by recycling some of the heat used in the evaporation process. Despite this, the greatest disadvantage of distillation is its high energy demand.

Figure 18 Hand held and operated **reverse osmosis** desalination units like this are suitable for producing potable water in a wide variety of situations. Note the lever handle used to pump and apply the pressure needed to initiate reverse osmosis. Image courtesy of Katadyn Products Inc.

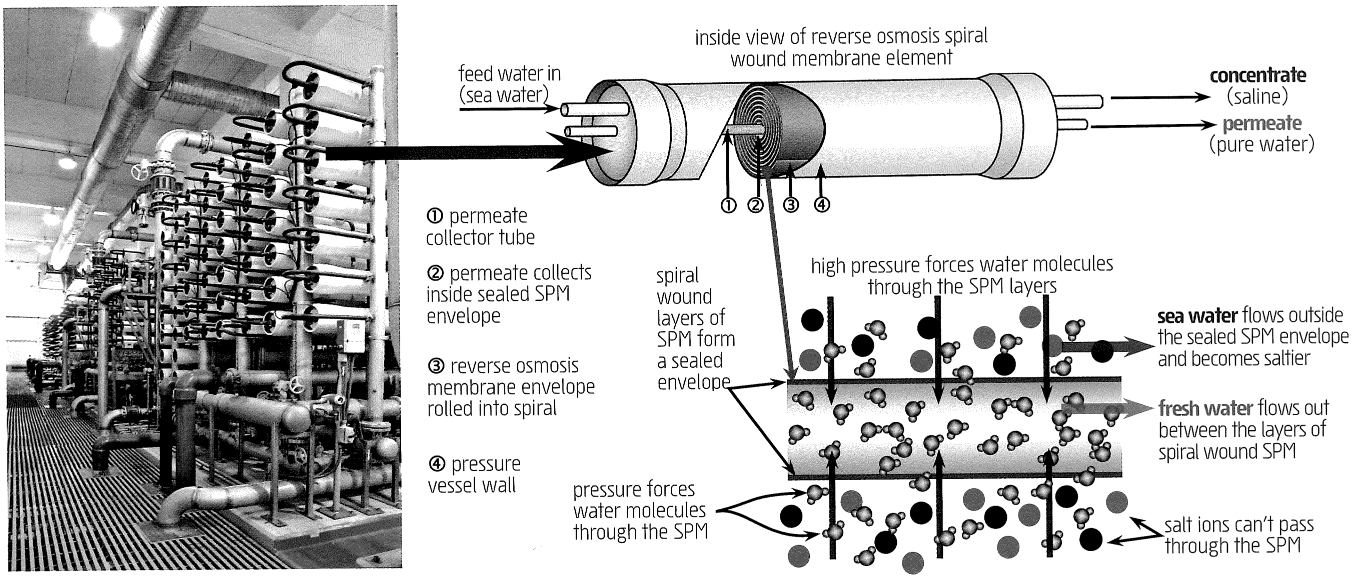


Another approach to obtaining fresh water from sea water involves a natural process, called osmosis, that plants use to draw water from the soil into their roots. The cell membrane of plant root cells acts like a **semi permeable membrane** (**SPM**) that allows water molecules to pass through but not larger molecules or salt ions. **Osmosis** is the natural tendency of water to diffuse through a SPM from a solution of low salt concentration to one of a higher salt concentration until the salt concentration on both sides of the SPM is the same. Thus plant root hair cells are able to absorb water through their cells SPM because the salt concentration inside the cells is maintained at a higher level than in the surrounding fresh water. As a result pure water is drawn into the cell's saltier interior by osmosis. If plants are watered with salty water they wilt because osmosis causes water to flow out of their root cells into the saltier surrounding water.

Commercial **reverse osmosis** (**RO**) units use high pressure to force water through a synthetic semi permeable membrane (**SPM**) in the opposite direction to which water would naturally flow. In this process pressure is applied to salty water (eg sea water), called the feed water, that flows on one side an SPM. This pressure causes water to flow out of the salty feed water and through the SPM to produce pure water. During RO the feed water becomes saltier as around 45% or more of the water content is removed. The resulting fresh water is called the permeate. (See border note.) Ions, larger organic molecules, cellular organisms and viruses do pass through the SPM.

Permeate is the term for a substance that diffuses through a semi permeable membrane. By comparison, **filtrate** is the term for a substance that passes through the microscopic pores of a filter paper during filtration.

Figure 19 The picture below shows a bank of cylindrical **reverse osmosis pressure vessels** (**PVs**) in a typical desalination plant. Each PV contains several spiral wound membrane elements (SWMEs) connected end on end down the length of the PV tube. Within each SWME fresh water, called permeate is separated from salty feed water (sea water). The SWME consists of several layers of material including a semi permeable membrane (SPM) in the form of a **rolled sealed envelope** plus various channel spacer layers that allow feed water (salty) and permeate water (fresh) to flow between the space on either side of the SPM. **High pressure** forces water molecules through the SPM and into the SPM envelope. Salt ions and other larger particles such as viruses and pathogenic organisms can't pass through the membrane.



In Western Australia, reverse osmosis (RO) is used to produce fresh water at both the Kwinana and Binningup desalination plants. At both plants permeate water (fresh water) is collected then treated for **disinfection** with chlorine as well as being **fluoridated** before storage or addition to the IWSS. Also, as the water produced by RO is essentially deionised water (distilled water) it is necessary to add some salts for both taste and health reasons. This is done by adding both carbon dioxide gas and lime (CaO) and results in a suitable pH as well as a total dissolved salt level of around 200 mg L^{-1}.

Although **energy consumption** is a significant issue in all desalination processes, the technology of reverse osmosis, especially the design of the membranes themselves, have greatly improved in the last decade resulting in significantly reduced power needs. Pressure exchange technology further enhances the energy efficiency of RO. This technology is used to transfer the residual high pressure of outgoing salty wastewater to incoming sea water. Reclaiming pressure this way can reduce overall RO plant energy requirements by 50-60%.

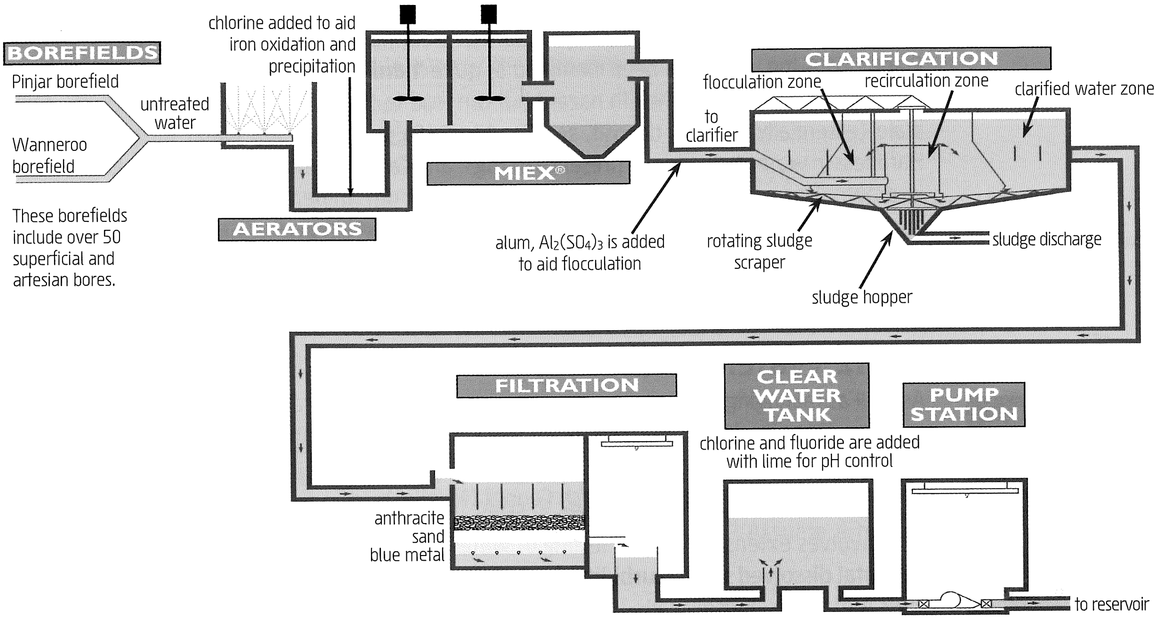
Typically reverse osmosis requires around a third the energy input of the most efficient thermal distillation plants. Also, for reasons of **sustainability** and **reduced greenhouse gas emissions** the Water Corporation (of WA) has offset the energy requirements of the Southern Seawater Desalination Plant by purchasing the entire energy output (≈65 MW) of two purpose built renewable energy farms near Geraldton.

Attempt Set 28 # 14 and 15.

16.11 Groundwater treatment

Groundwater currently supplies about 40% of the needs for the IWSS which supplies Perth, the South West, the Goldfields and Agricultural regions. At present around 180 bores are used to draw water from the Yarragadee and Leederville aquifers. Most of these bores are located in Perth's northern suburbs. Groundwater is treated at six groundwater treatment plants to remove contaminants that may include finely suspended solids as well as various dissolved substances like manganese, iron, hydrogen sulfide, carbon dioxide and organic compounds.

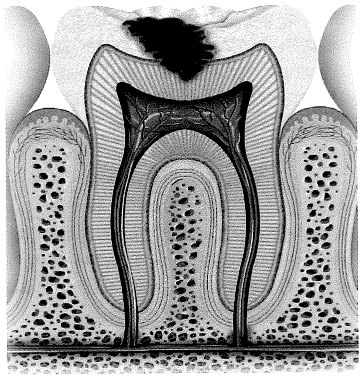
Figure 20 The **Wanneroo groundwater treatment plant** features an Australian developed **MIEX®1** Resin process (magnetised ion exchange) that prevents an intermittent 'swampy' odour occurring in treated groundwater supplied to Perth's northern suburbs. Unlike conventional ion exchange, the MIEX®1 resin more effectively removes dissolved organic compounds from drinking water, the source of the odour and taste concerns. Image courtesy of the Water Corporation of WA.



Some of the typical processes involved in groundwater treatment include:

- **Aeration:** Groundwater often contains dissolved gases like carbon dioxide (CO_2) and hydrogen sulfide (H_2S). These increase the acidity of water while H_2S also gives water a bad odour. Spraying groundwater into the air helps to expel these gases and increase the concentration of dissolved oxygen. The extra dissolved oxygen, along with some added chlorine, oxidise and remove any dissolved organic compounds that may be present. The added Cl_2*(aq)* and O_2*(aq)* also reduce the solubility of dissolved iron and manganese. This assists in their later removal by clarification and filtration. While neither iron or manganese are toxic, manganese compounds cause water to have a black colouration and iron compounds give water a brown appearance. Both cause staining of clothing and other surfaces.
- **Clarification:** This occurs after aeration and involves the removal of fine particles that may be suspended in the groundwater. Such particles will not normally settle out and their presence causes water to be turbid (unclear) and coloured. Clarification is achieved in a large settling tank where alum, $Al_2(SO_4)_3$ and lime, CaO are added while stirring. Alum is a coagulant or flocculating agent. It causes fine suspended particles to clump together and quickly settle out. Clarified water is then drawn off the top of the settling tanks.
- **Sand filtration:** Clarified groundwater may still be coloured and contain fine suspended particles. To remove these the water is passed through filter beds of granulated anthracite (coal with up to 98% carbon) and coarse sand. Anthracite is very effective at adsorbing organic materials that otherwise cause water to be coloured.
- **Disinfection:** Despite clarification and filtration, harmful pathogenic bacteria and viruses may be present. Chlorination or chloramination is used to destroy these. The latter involves the use of chlorine in conjunction with ammonia to produce a longer lasting disinfectant. This is particularly relevant to the Goldfields and Agricultural Water Supply Scheme where disinfection is required across an extensive and long network of pipes. Sufficient chlorine is added to ensure its concentration stays slightly above 1 ppm at the consumer outlet.
- **Fluoridation:** The presence of fluoride ions in drinking water is known to reduce tooth cavity rates by strengthening tooth enamel. (See Fig 21.) To maximise the public health benefit from fluoridation the fluoride concentration in drinking water supplies in Western Australia are maintained between 0.6-1.0 mg L^{-1} (0.85 mg L^{-1} is considered ideal). A maximum dose rate of 1 mg L^{-1} is prescribed in the 'Fluoridation of Public Water Supplies Act' which is managed by the Department of Health. Fluoridation involves adjusting the natural fluoride concentration often present in groundwater to between 0.6 mg L^{-1} and 1.0 mg L^{-1} by adding fluorosilicic acid, (H_2SiF_6). In WA this is obtained as a by-product of phosphate fertiliser manufacture. These levels are regularly monitored at each of the individual treatment plants. (See border note.)

Figure 21 Fluoride is a naturally occurring ion found in many water sources. Its concentration may vary from less than 0.1 mg L^{-1} in surface waters through to greater than 1.5 mg L^{-1} in some groundwater sources. Health authorities have long recognised that fluoride levels of up to 1.5 mg L^{-1} reduce tooth cavity rates by strengthening tooth enamel. This occurs as fluoride changes the mineral in tooth enamel from **hydroxyapatite**, $Ca_5(PO_4)_3OH$*(s)* into **fluorapatite**, $Ca_5(PO_4)_3F$*(s)*, a more acid resistant mineral.



To maximise the public health benefit from fluoridation the fluoride concentration in drinking water supplies in Western Australia is maintained between **0.6-1.0 mg L^{-1}**. To confirm acceptable fluoridation, the fluoridated water supplies are sampled and tested at least weekly.

It is also known that constant exposure to higher fluoride levels (greater than 1.5 mg L^{-1}) may lead to **dental fluorosis**, an aesthetic mottling of tooth enamel. Very high fluoride exposure can lead to a skeletal problem called skeletal fluorosis.

With this in mind water fluoridation is supported by the World Health Organisation, the Australian Dental Association, the Australian Medical Association and the National Health Medical and Research Council.

- **pH:** Australian guidelines specify an aesthetic pH range of 6.5 to 8.5. pH correction is achieved by the addition of lime, CaO (for increasing pH) or by dissolving carbon dioxide gas, CO_2 (lowers pH). Stronger acids and bases such as sulfuric acid or sodium hydroxide may also be used for this purpose.

In some areas groundwater may require further treatment. At the Neerabup Water Treatment Plant in WA the water has been identified as quite **'hard'**, meaning it will not easily lather with soap. Hard water is not a health hazard but can lead to the build up of mineral deposits, called scale (essentially a $CaCO_3$ deposit, see Fig 12 p140), in kettles, hot water pipes and fittings. Hardness in water is due to the presence of $\mathbf{Mg^{2+}}$ and $\mathbf{Ca^{2+}}$ ions and can occur when groundwater is drawn from areas rich in calcium or magnesium minerals. The problem has been addressed at the Neerabup Water Treatment Plant by using a crystallisation process that rapidly precipitates Ca^{2+} ions as calcium carbonate onto seed crystals of calcite ($CaCO_3$).

16.12 Water monitoring

As well as managing the water treatment process chemists play an important role in monitoring water quality. The Water Corporation of WA for example, in conjunction with the Department of Health, manage water quality consistent with Australian guidelines as specified by the National Health and Medical Research Council (NHMRC).

This involves extensive regular testing of water quality for a number of properties including **pH**, **total dissolved solids**, **turbidity**, **bacterial** and **viral pathogens** plus a range of potentially toxic metal ions, loosely called **heavy metals**, including, Sb, Cd, Cr, Cu, Al, Pb, Ni, Zn, As, Ba, Be, B, Hg, Mo, Se, Ag and U, as well as a variety of hydrocarbons, pesticides, nitrates (NO_3^-), cyanides (CN^-) and iodides (I^-). In 2011-2012 the Water Corporation in WA took more than 65,000 water samples and produced in excess of 275,000 individual analyses on these samples.

Arsenic monitoring is particularly important for bores in the Gwelup region (11 km north of Perth) where very high concentrations of arsenic, iron and aluminium have recently been identified in near surface water. A study of **domestic garden bores** in the Gwelup area in 2004 identified 50 bores with pH<5.5. Of these bores, dissolved iron was recorded at concentrations up to 1300 mg L^{-1}, aluminium up to 290 mg L^{-1}, and arsenic up to 0.8 mg L^{-1}. While there is no health guideline for iron or aluminium, for arsenic this represents a concentration eighty times the NHMRC guideline of 0.01 mg L^{-1}. Other tests on water from purpose drilled bores in the area have since recorded arsenic concentrations of up to 7.3 mg L^{-1} at shallow depths near the water table. It is **important to note** that water used for public consumption is drawn from much deeper levels in the Gnangara Mound where such contamination has not been recorded. It is important however, to regularly monitor groundwater for the presence of such potential contaminants. (See Fig 22.)

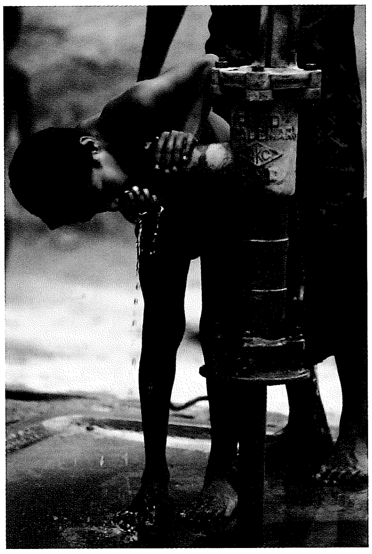


Figure 22 Wells like this **shallow tube well** became a very important water source in Bangladesh from the late twentieth century. This followed an extensive shallow tube well drinking water program intended to provide villagers a bacteria free water supply for drinking washing and agriculture. Unfortunately authorities did not test for **arsenic** in the groundwater and it is now known that many of these shallow wells are heavily contaminated with naturally occurring arsenic. Similar problems are known to occur in many parts of **Southeast Asia**, where similar hydrology results in high concentrations of arsenic in near surface groundwater.
Some effects the villagers suffer as a result of exposure to arsenic in their drinking water include skin rashes and thickening, nausea, diarrhoea and vomiting, partial paralysis, numbness in hands and feet and blindness. Longer term exposure is associated with a variety of cancers such as lung, kidney, liver, prostate, bladder and skin cancer.

Complete Set 28.

Set 28 Solution concentration and drinking water

1. What is the **concentration** in mol L^{-1} of each solute in the following solutions?
   a. a 2.40 L solution of NaOH*(aq)* containing 1.41 mol of NaOH
   b. a 561 mL sample of vinegar solution containing 0.71 mol of CH_3COOH

2. What is the **molarity** (concentration in mol L^{-1}) of the solutes in the following solutions?
   a. 16.2 g of $CuSO_4$ dissolved in 1.94 L of solution
   b. 139 mg of NaCl dissolved in 0.494 L of solution
   c. 11.77 L of solution containing 129 g of iron(III) sulfate-9-water ($Fe_2(SO_4)_3.9H_2O$)